Other Recurrent Units

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Long Short-Term Memory (LSTM)

Cell state c_t - help decide what info to keep/forget

Forget gate f_t - control what information from c_{t-1} is forgotten

Input gate i_t - what new info should be added to c_t from x_t

Output gate q_t - how h_t should be updated given c_t

Candidate cell state $\tilde{c}_t$ - estimate of cell state using only h_{t-1}, x_t

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

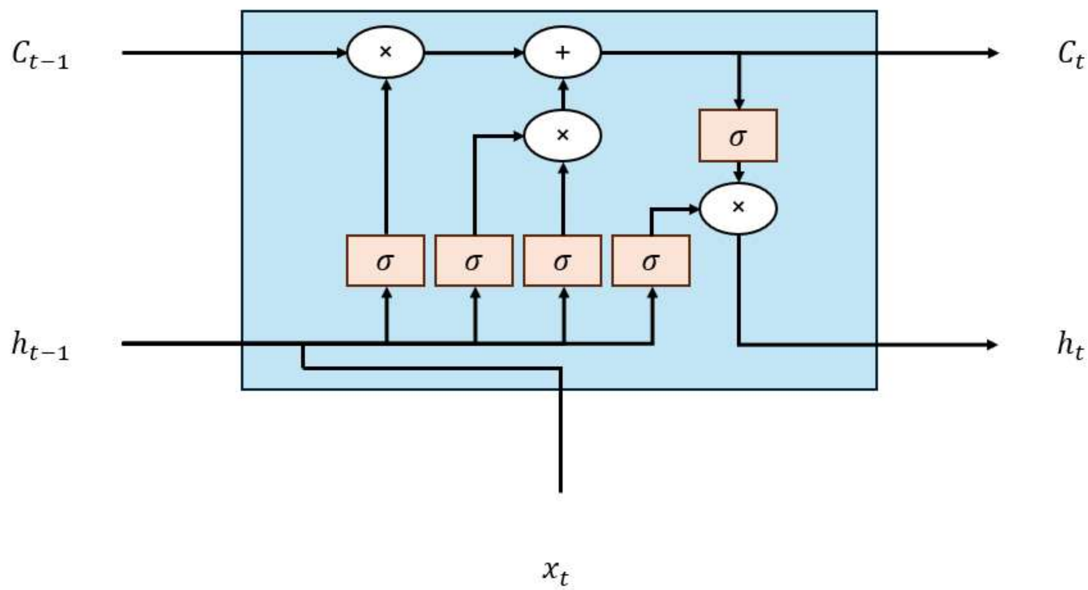
$$h_t = q_t \odot \sigma(c_t)$$

$$f_t = \sigma(b_f + W_f h_{t-1} + U_f x_t)$$

$$i_t = \sigma(b_i + W_i h_{t-1} + U_i x_t)$$

$$q_t = \sigma(b_q + W_q h_{t-1} + U_q x_t)$$

$$\tilde{c}_t = \sigma(b_c + W_c h_{t-1} + U_c x_t)$$



Gated Recurrent Unit

Update gate u_t - what info from previous hidden state to keep/forget

Reset gate r_t - what info from previous hidden state to forget

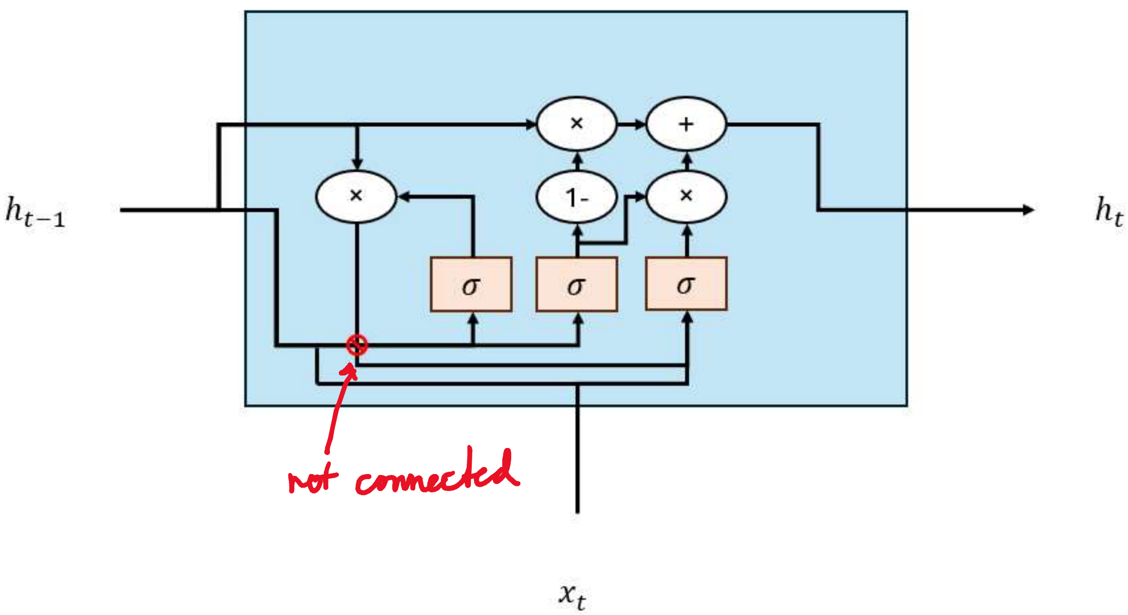
Candidate hidden state $\tilde{h}_t$ - estimate of updated hidden state with flow controlled by reset gate by reset gate

$$h_t = (1 - u_t) \odot h_{t-1} + u_t \odot \tilde{h}_t$$

$$\tilde{h}_t = \sigma(b_h + W_h(r_t \odot h_{t-1}) + U_h x_t)$$

$$u_t = \sigma(b_u + W_u h_{t-1} + U_u x_t)$$

$$r_t = \sigma(b_r + W_r h_{t-1} + U_r x_t)$$



Vanilla	LSTM	GRU
Short	Long	Long
Simple	More parameters	Fewer parameters
	Expressive power	More efficient

Simple	Expressive power Large dataset Complex tasks	More efficient than LSTM